A2 Review of 6.1/6.2 Simplifying, multiplying and dividing

MULTIPLYING RADICALS. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$

Examples. Multiply and **express in simplest radical form**. All variables are > 0.

1. $\sqrt[2]{6} \cdot \sqrt[2]{3}$

$\sqrt[2]{18}$ $\boxed{3\sqrt{2}}$

9 2

3 3

2. $\sqrt[3]{4} \cdot \sqrt[3]{2}$

$\sqrt[3]{8} = \boxed{2}$

4 2

2 2

3. $\sqrt[3]{6} \cdot \sqrt[3]{2}$

$\sqrt[3]{12}$ $\boxed{\sqrt[3]{12}}$

3 4

2 2

720

12 60

6 2 10 6

5 2

4. $\sqrt[2]{12x^3} \cdot \sqrt[2]{60x^9}$

x x x x x x x x x x x x

$\sqrt{720x^{12/2}}$

36 20

6 6 4 5

2 2

$6 \cdot 2x^6\sqrt{5} = \boxed{12x^6\sqrt{5}}$

5. $3\sqrt{5xy} \cdot 2\sqrt{15xy^2}$

$6\sqrt{75x^2y^3}$

25 3

5 5

$x^{2/2}$ $y^{2/2} \cdot y^1$

$6 \cdot 5xy\sqrt{3y}$

$\boxed{30xy\sqrt{3y}}$

6. $4\sqrt[3]{-9xyz^2} \cdot 3\sqrt[3]{-6xy^2z^5}$

$12\sqrt[3]{54x^2y^3z^7}$

9 6

3 3 3 2

$y^{3/3}$ $z^{6/3}$ z^1

$12 \cdot 3yz^2\sqrt[3]{2x^2z}$

$\boxed{36yz^2\sqrt[3]{2x^2z}}$

DIVIDING RADICALS

$$\boxed{\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}}$$

REMEMBER : **NO RADICALS left IN THE DENOMINATOR** All variables are > 0.

1. $\frac{\sqrt{108}}{\sqrt{2}} = \sqrt{\frac{108}{2}}$

$= \sqrt{54}$

9 6

3 3 3 2

$\boxed{3\sqrt{6}}$

2. $\frac{\sqrt{12}}{\sqrt{20}}$

$= \sqrt{\frac{12 \div 4}{20 \div 4}}$

$\sqrt{\frac{3}{5}}$

$\frac{\sqrt{3}\cdot\sqrt{5}}{\sqrt{5}\cdot\sqrt{5}} = \frac{\sqrt{15}}{\sqrt{25}} = \boxed{\frac{\sqrt{15}}{5}}$

3. $\frac{7}{\sqrt{15a^2}}$ $a^{2/2}$

$\frac{7\sqrt{15}}{a\sqrt[2]{15}\cdot\sqrt{15}} = \frac{7\sqrt{15}}{a\sqrt{225}}$

$= \boxed{\frac{7\sqrt{15}}{15a}}$

4. $\frac{5}{\sqrt{2x^3}}$

$x^{2/2}\,x^1$

$\frac{5\cdot\sqrt{2x}}{x\sqrt{2x}\cdot\sqrt{2x}}$

$\frac{5\sqrt{2x}}{x\cdot 2x} = \boxed{\frac{5\sqrt{2x}}{2x^2}}$

5. $\frac{y}{\sqrt{36y^5}}$

6 6 $y^{4/2}\,y^1$

$\frac{y\cdot\sqrt{y}}{6y^2\sqrt{y}\cdot\sqrt{y}} = \frac{\cancel{y}\sqrt{y}}{6y^2\cdot\cancel{y}} = \boxed{\frac{\sqrt{y}}{6y^2}}$

6. $\frac{\sqrt{3x^3y^2}}{\sqrt{6x^4y^4}}$

$y^{2-4} = y^{-2}$

$x^{3-4} = x^{-1}$

$\frac{1}{xy^2}$

$\cancel{x}\cancel{x}\cancel{x}$ $\cancel{y}\cdot\cancel{y}$

$\sqrt{\frac{3x^3y^2}{6x^4y^4}}$

$\cancel{x}\cancel{x}\cancel{x}x$ $\cancel{y}\cdot\cancel{y}\cdot y\cdot y$

$\sqrt{\frac{1}{2xy^2}} = \frac{\sqrt{1}}{\sqrt{2xy^{2/2}}} = \frac{1\cdot\sqrt{2x}}{y\sqrt{2x}\cdot\sqrt{2x}}$

$= \boxed{\frac{\sqrt{2x}}{2xy}}$